Spherical Cyclic Time

A Reduced Informational Toy Framework for Relative Restart, Causal-Patch Capacity, and Perturbative Memory

Version 1.4-H - Referee-Response and Prepublication Patch

Pietro Risi

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Editorial status. This document updates the English v1.2.0 working paper using the v1.3-A to v1.4-H audit line, including the prepublication referee-response patch. It is not peer reviewed. It does not claim to establish a physical cyclic cosmology, a nonsingular bounce, a new gravitational theory, an observational signature, or a microphysical derivation of the restart mechanism. Its intended status is a controlled toy/informational framework with explicit failure classes and a reduced mainline.

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Keywords: cyclic time; spherical time; entropy; causal patch; apparent horizon; informational capacity; relative restart; cyclic cosmology; phenomenological model; perturbative memory; holographic bound.

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Abstract

This paper presents *Spherical Cyclic Time* (SCT), a reduced toy/informational framework for relative restart under a non-decreasing absolute entropy floor. The framework separates cyclic phase, $\phi \in S^1$, from unwrapped physical time, $\tau \in \mathbb{R}$, so that return of phase is not identified with return of a physical microstate. Absolute entropy is written as

$$S_{\text{abs}}(n, u) = S_{\text{min}}(n) + \Delta S_n(u),$$

with $S_{\text{min}}(n+1) \geq S_{\text{min}}(n)$, while the internal relative entropy is $S_{\text{rel}}(n, u) = \Delta S_n(u)$. A restart is therefore modeled only as a relative operational reset, not as an absolute entropy erasure.

The v1.3 update reduces the model to a conservative mainline. The label "Spherical is retained as a historical visualization label only: the auxiliary sphere does not carry the physical or mathematical content of the framework. The accessible capacity is treated conceptually as $S_{\text{cap}} = \ln N_{\text{acc}}$, constrained by a causal-patch area bound. The mainline operational choice is the flat-FLRW apparent/Hubble patch, giving the normalized capacity

$$\widehat{S_{\text{cap}}}(u) = E^{-2}(u), \qquad E^2(u) = 1 + C_m e^{-3u} + C_r e^{-4u}.$$

The future event horizon is retained only as a patch-sensitivity diagnostic, even though it gives more favorable toy classifications in the scan.

The entropy-production parameter is rewritten as $\alpha_{\text{eff}}(u) = \Gamma_S(u)/H(u)$. A fixed physical-clock toy candidate, $\Gamma_S = \gamma_\infty H_\infty$, is retained as the least problematic template from the v1.3-B scan, but γ_∞ is not derived. The restart diagnostic remains

$$q_{\text{eff}}(u) = \frac{\Delta S_n(u)}{S_{\text{cap}}(u)},$$

with the crossing $q_{\text{eff}} \geq q_{\text{crit}}$ interpreted as phenomenological classification only.

For perturbative continuity, the update uses

$$\delta_{n+1}(k) = T_R(k)\delta_n(k) + \xi_n(k), \qquad P_{n+1}(k) = |T_R(k)|^2 P_n(k) + P_\xi(k),$$

and the memory fraction $F_{\text{mem}}(k)$. The cautious mainline is partial large-scale memory with small-scale regeneration, represented by a soft filter such as $T_R(k) = \exp[-(k/k_c)^p]$. The parameters k_c, p, and P_ξ are not derived.

The conclusion is deliberately narrow: SCT is not a completed cyclic cosmology. It is a grammar for testing whether relative restart can be represented coherently while preserving failures, patch sensitivities, and novelty boundaries.

1 Methodological rule: exploration without forcing

This work follows a fixed rule: parameters and mechanisms must not be selected after the fact merely to obtain clean outcomes. If a branch works in a limited toy region, that is recorded. If it fails, becomes source-dominated, restarts too early, or depends on an unresolved patch choice, that is also recorded.

In practical terms:

1. no parameter is chosen only because it makes a graph look successful;
2. failed regions remain visible in tables and appendices;
3. toy scans are not presented as cosmological evidence;
4. a negative result is retained as part of the model audit;
5. the final output may be only a map of viable and non-viable toy structures.

Operational principle: explore, do not force. A mechanism that only works by hidden tuning, source domination, or post-hoc patch selection is not promoted to the mainline.

2 v1.4-H prepublication referee-response patch

A referee-style external review of the v1.3/v1.4 package returned a narrow positive verdict for the toy/informational mandate, while identifying minor issues that could make the paper look more physical than intended. This v1.4-H patch therefore applies editorial and scientific guardrails before Zenodo publication. The patch is not a new physical model. It does four things:

1. marks the spherical construction as a historical auxiliary visualization label;
2. strengthens the hedge on $\widehat{S_{\mathrm{cap}}} = H_\infty^2/H^2$;
3. labels all scan tables as deterministic toy-classifier counts, not empirical evidence;
4. records threshold sensitivity of the v1.3-B classifier without optimizing thresholds.

3 Scope and claim boundary

The present framework is not a new gravitational theory, not a complete cyclic universe, not a bounce model, and not an alternative to inflation. It also does not derive the Bousso bound, de Sitter entropy, or Penrose's conformal cyclic cosmology. The intended claim is narrower:

SCT is a toy/informational grammar for relative restart under a monotone absolute entropy floor, using a causal-patch capacity diagnostic and a perturbative memory/regeneration classifier.

The possible value of the framework lies in a small set of controlled distinctions:

- cyclic phase is not physical recurrence;
- relative entropy can be reset without resetting absolute entropy;
- accessible capacity can act as an internal diagnostic;

- restart can be classified by capacity saturation rather than assumed;
- perturbative memory can be separated from small-scale regeneration;
- failure classes are part of the result.

4 Auxiliary spherical timeline geometry

The original SCT visualization represents possible timelines as great circles on an auxiliary sphere. Let

$$S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}, \qquad p = e_3 = (0, 0, 1).$$

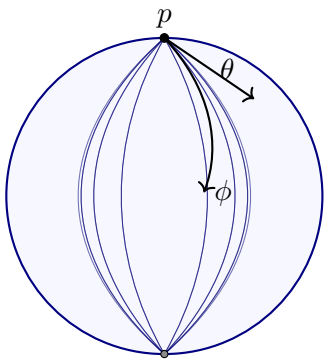
In the tangent plane at p, a direction is

$$u(\theta) = (\cos\theta, \sin\theta, 0), \qquad \theta \in [0, 2\pi).$$

Each θ defines a great circle parametrized by the cyclic phase ϕ:

$$\gamma_\theta(\phi) = \cos\phi\, e_3 + \sin\phi\, u(\theta), \qquad \phi \in [0, 2\pi).$$

Thus $\gamma_\theta(0) = \gamma_\theta(2\pi) = p$.



family of timelines as great circles

Figure 1: Auxiliary representation of timelines as great circles through a common point. The variable θ labels a timeline; ϕ labels cyclic phase.

This sphere is not a Lorentzian spacetime metric. It is an auxiliary diagram and a historical visualization label only. The word “Spherical” in the title should not be read as a claim that the framework contains a spherical spacetime geometry or that the auxiliary sphere performs physical work. The physical content of the toy framework is in the entropy bookkeeping, capacity diagnostic, restart classification, and perturbative matching grammar.

5 Cyclic phase and unwrapped time

The reduced mainline keeps the separation

$$\phi \in S^1, \qquad \tau \in \mathbb{R}.$$

A simple representation is $\phi(\tau) = \tau \pmod{2\pi}$. The important point is negative: returning to the same ϕ is not a return to the same microstate. Entropy floors, perturbative memories, and accumulated backgrounds can continue to evolve with the unwrapped variable τ and cycle label n.

6 Relative restart and entropy bookkeeping

Let n label the cycle and let u be a monotone internal variable, often $u = \ln(a/a_0)$. The entropy split is

$$S_{\text{abs}}(n,u) = S_{\text{min}}(n) + \Delta S_n(u), \tag{1}$$

with relative entropy

$$S_{\text{rel}}(n,u) = S_{\text{abs}}(n,u) - S_{\text{min}}(n) = \Delta S_n(u). \tag{2}$$

The absolute floor obeys

$$S_{\text{min}}(n+1) \geq S_{\text{min}}(n). \tag{3}$$

A relative restart may therefore have $S_{\text{rel}}(n+1,0) \approx 0$ without $S_{\text{abs}} \to 0$.

At a restart, schematically,

$$S_{\text{min}}(n+1) = S_{\text{min}}(n) + \Delta S_{\text{cycle}}(n), \qquad \Delta S_{\text{cycle}}(n) = \Delta S_n(u_*). \tag{4}$$

This is why the restart is relative: the internal zero is renormalized, while the absolute floor is not erased.

7 Accessible capacity and causal-patch choice

At the conceptual level, accessible informational capacity is

$$S_{\text{cap}}(t) \coloneqq \ln N_{\text{acc}}(t), \tag{5}$$

where N_{acc} is the effective number of physically distinguishable states accessible inside the relevant patch. At the bound-motivated level,

$$S_{\text{cap}}(t) \lesssim \frac{A_{\text{patch}}(t)}{4G}. \tag{6}$$

The v1.3-A audit fixes the operational mainline to the flat-FLRW apparent/Hubble patch,

$$R_A = \frac{1}{H}, \qquad A_A = \frac{4\pi}{H^2}. \tag{7}$$

After normalization to the asymptotic quasi-de Sitter plateau,

$$\widehat{S_{\text{cap}}} = \frac{H_\infty^2}{H^2}. \tag{8}$$

With

$$H^2(u) = H_\infty^2 \left(1 + C_m e^{-3u} + C_r e^{-4u}\right), \tag{9}$$

this becomes

$$E^2(u) = 1 + C_m e^{-3u} + C_r e^{-4u}, \qquad \widehat{S_{\text{cap}}}(u) = E^{-2}(u). \tag{10}$$

The H^{-2} capacity is retained only as an apparent/Hubble-patch normalized area diagnostic in flat FLRW. It is not a derivation of the true cosmic microstate count, not a computation of N_{macc}, and not a full application of the covariant entropy bound. The covariant bound requires a surface and non-expanding light-sheets; SCT does not construct those objects here.

The future event horizon was tested as a diagnostic alternative. It produced more favorable toy counts in the v1.3-A scan, but it is future-dependent and is not selected merely because it improves the classification.

Table 1: v1.3-A patch-choice sensitivity counts. These are deterministic toy-classifier bin counts over an explicitly specified grid; they are not measurements, observational evidence, or cosmological parameter estimates.

Patch	**Classification**	**Count**
Apparent/Hubble mainline	ok toy	216
Apparent/Hubble mainline	failed: no restart within $u_{\max}$	240
Apparent/Hubble mainline	critical: early restart	196
Apparent/Hubble mainline	weak: late scale	49
Apparent/Hubble mainline	weak: immature capacity	11
Apparent/Hubble mainline	critical: very immature capacity	8
Future event horizon diagnostic	ok toy	330
Future event horizon diagnostic	failed: no restart within $u_{\max}$	240
Future event horizon diagnostic	critical: early restart	45
Future event horizon diagnostic	weak: late scale	40
Future event horizon diagnostic	weak: immature capacity	43
Future event horizon diagnostic	critical: very immature capacity	22

8 Entropy-production clock: from α to Γ_S/H

The v1.2.0 dynamic model used

$$\frac{d\Delta S}{du} = \alpha\left(1 - \frac{\Delta S}{S_{\text{cap}}}\right), \qquad \Delta S(0) = 0. \tag{11}$$

Since $du/dt = H$, the same equation can be written as

$$\frac{d\Delta S}{dt} = \Gamma_S(u)\left(1 - \frac{\Delta S}{S_{\text{cap}}}\right), \qquad \alpha_{\text{eff}}(u) = \frac{\Gamma_S(u)}{H(u)}. \tag{12}$$

This reinterpretation does not derive Γ_S. It only prevents α from remaining a purely formal knob.

The v1.3-B scan compared several diagnostic templates using the same broad amplitude grid:

$$\alpha_{\text{legacy}}(u) = \alpha_0, \tag{13}$$
$$\alpha_{\text{clock}}(u) = \gamma_\infty E^{-1}(u), \tag{14}$$
$$\alpha_{\text{cap}}(u) = \lambda \frac{d \ln S_{\text{cap}}}{du}, \tag{15}$$
$$\alpha_m(u) = \lambda \Omega_m(u), \tag{16}$$
$$\alpha_r(u) = \lambda \Omega_r(u), \tag{17}$$
$$\alpha_{m\Lambda}(u) = 4\lambda \Omega_m(u)\Omega_\Lambda(u). \tag{18}$$

The least problematic retained toy candidate is the fixed physical-clock template

$$\Gamma_S = \gamma_\infty H_\infty, \qquad \alpha_{\text{eff}}(u) = \frac{\gamma_\infty}{E(u)}. \tag{19}$$

It suppresses early production when H is large without adding an explicit transition time by hand. However, γ_∞ remains unresolved and is not microphysically derived.

Table 2: Condensed v1.3-B source-template outcomes. These are deterministic toy-classifier bin counts over an explicitly specified grid; they are not measurements, observational evidence, or cosmological parameter estimates. The counts are not used to tune a preferred amplitude.

Template	**ok**	**failed**	**weak/no**	**weak**	**critical**	**total**
Fixed physical clock $\Gamma_S = \text{const}$	345	196	44	133	2	720
Legacy constant α	333	195	45	127	20	720
Capacity-growth locked	0	600	60	5	55	720
Matter fraction window	0	708	12	0	0	720
Radiation fraction window	0	720	0	0	0	720
Matter–Λ transition	3	632	47	28	10	720

The important result is not that one template can be made to work. The important result is that many physically suggestive templates fail in the toy scan and are therefore not promoted to the mainline.

9 Restart diagnostic and internal classification

The effective saturation diagnostic is

$$q_{\text{eff}}(u) = \frac{\Delta S(u)}{S_{\text{cap}}(u)}. \tag{20}$$

A phenomenological restart is recorded when

$$q_{\text{eff}}(u_*) \geq q_{\text{crit}}. \tag{21}$$

The crossing is a classifier, not a derived law of nature. A case is considered stronger if the capacity has matured, for example

$$S_{\text{cap}}(u_*) \geq S_{\text{mature}}. \tag{22}$$

In the baseline tables, the diagnostic thresholds are fixed at $S_{\text{mature}} = 0.9$ and $1 \leq u_* \leq 12$ for an "ok toy" classification. These thresholds are not derived physical constants. They are classifier settings. A v1.4-H prepublication audit varied $S_{\text{mature}} \in \{0.8, 0.9, 0.95\}$, $u_{\min} \in \{0.5, 1.0, 2.0\}$, and $u_{\max} \in \{8, 12, 16\}$ using the same v1.3-B dynamic outputs. The result is a threshold-dependency audit, not an optimization.

Table 3: v1.4-H prepublication threshold-sensitivity audit for the v1.3-B classifier. The dynamics are not rerun or tuned; only fixed diagnostic thresholds are varied.

Quantity	**min**	**median**	**max**
Total ok toy	486	639	805
Fixed physical-clock ok	246	322	405
Legacy constant-alpha ok	239	316	390
Matter-Lambda transition ok	1	3	9

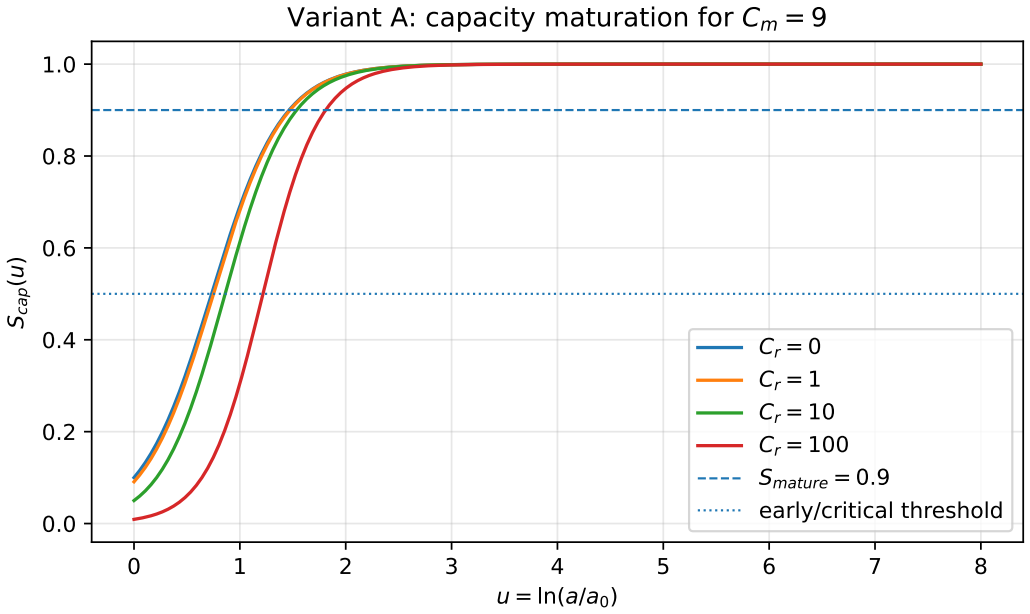


Figure 2: Capacity maturation in the v1.2/v1.3 apparent-Hubble capacity branch. A restart before sufficient capacity maturation is not counted as a full success.

The working classes are:

- **coherent toy**: restart appears after capacity maturation, without extreme scale or source domination;
- **weak**: restart exists but is late, immature, or parameter-sensitive;
- **critical**: restart is too early, too extreme, or structurally suspicious;
- **failed**: no restart appears in the scanned window or consistency is violated;
- **suspicious rescue**: an auxiliary source creates the restart by dominating the dynamics.

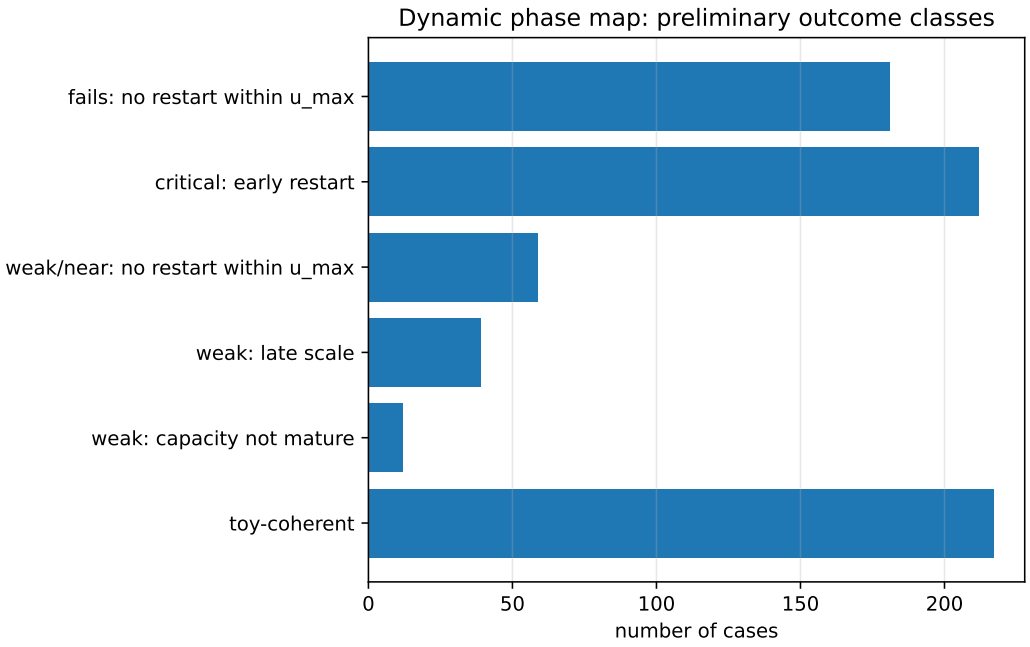


Figure 3: Baseline v1.2.0 dynamic phase-map count. The update keeps failure regions visible rather than optimizing them away.

10 Perturbative matching across relative restart

A cyclic framework must specify what passes through restart. The minimal effective grammar is

$$\delta_{n+1}(k) = T_R(k)\delta_n(k) + \xi_n(k), \tag{23}$$

where $T_R(k)$ is a restart-transfer operator and $\xi_n(k)$ is a regenerated component. At the power-spectrum level,

$$P_{n+1}(k) = |T_R(k)|^2 P_n(k) + P_\xi(k), \tag{24}$$

assuming no cross-correlation. Define

$$F_{\mathrm{mem}}(k) = \frac{|T_R(k)|^2 P_n(k)}{|T_R(k)|^2 P_n(k) + P_\xi(k)}. \tag{25}$$

The intended toy regime is not total memory. The cautious mainline is partial memory at large scales and regeneration at small scales. A soft diagnostic transfer is

$$T_R(k) = \exp\left[-\left(\frac{k}{k_c}\right)^p\right]. \tag{26}$$

The parameters k_c, p, and P_ξ are not derived here.

The v1.3-C scan contained 5040 rows:

Table 4: v1.3-C perturbative matching classification counts. These are deterministic toy-classifier bin counts over an explicitly specified grid; they are not measurements, observational evidence, or cosmological parameter estimates.

Class	**Reason**	**Count**
Coherent toy	large-scale partial memory, small-scale regeneration	228
Critical	memory-dominant suspicious	1252
Critical	small-scale memory dominant	126
Failed	memory erased at all scales	784
Failed	no large-scale transmission	293
Weak	not a clean transition	1909
Weak	borderline large-scale memory	285
Weak	small-scale memory not cleanly erased	135
Weak	coherent shape but sharp-filter stress test	28

Only a minority of the scanned cases are coherent toy cases. This is a useful constraint, not a problem to hide.

11 Literature and novelty boundary

The v1.3-D audit narrows the novelty claim. SCT is not novel as cyclic cosmology, entropy-bound physics, conformal cyclic cosmology, bounce construction, or perturbation theory. The surrounding literature already includes Tolman-style entropy accumulation in cyclic universes, modern ekpyrotic/cyclic scenarios, black-hole accounting in cyclic settings, Penrose's conformal cyclic cosmology and Weyl curvature hypothesis, holographic/covariant entropy bounds, and de Sitter horizon thermodynamics.

SCT does not claim to:

- solve the Tolman entropy problem;
- derive the Bousso or covariant entropy bound;
- derive Gibbons–Hawking/de Sitter entropy;
- implement Penrose conformal cyclic cosmology;
- construct a nonsingular bounce;
- replace inflation;
- predict an observational signature;
- derive Γ_S, k_c, p, or P_ξ from microphysics.

The defensible novelty boundary is limited to the toy/informational organization: relative restart with a monotone entropy floor, causal-patch capacity as a diagnostic, and an explicit memory/regeneration classifier.

This positioning is especially important relative to dynamical ekpyrotic/cyclic cosmologies. In such models, reset-like behavior, entropy dilution, perturbation transfer, or bounce physics must be implemented by explicit dynamics. SCT does not replace those models; it abstracts one bookkeeping idea–relative restart under monotone absolute entropy–without deriving the required dynamics.

12 Reduced v1.3 mainline

After v1.3-A through v1.3-E, the mainline is:

1. auxiliary spherical representation for phase families only;
2. $\phi \in S^1$ and $\tau \in \mathbb{R}$;
3. $S_{\text{abs}}(n, u) = S_{\text{min}}(n) + \Delta S_n(u)$;
4. $S_{\text{min}}(n+1) \geq S_{\text{min}}(n)$;
5. $S_{\text{cap}} = \ln N_{\text{acc}} \lesssim A_{\text{patch}}/4G$;
6. flat-FLRW apparent/Hubble patch as operational mainline, $\widehat{S_{\text{cap}}} = E^{-2}$;
7. $\alpha_{\text{eff}} = \Gamma_S/H$;
8. fixed physical-clock toy candidate, $\Gamma_S = \gamma_\infty H_\infty$;
9. restart classifier $q_{\text{eff}} = \Delta S/S_{\text{cap}}$;
10. perturbative matching classifier $\delta_{n+1} = T_R \delta_n + \xi_n$;
11. soft transfer stress grammar $T_R(k) = \exp[-(k/k_c)^p]$.

Declassified to appendices or stress tests:

- future event horizon capacity;
- matter, radiation, matter–Λ, and capacity-growth source clocks;
- legacy constant-α scans;
- Weyl/black-hole dominated rescue branches from v1.2.0;
- sharp perturbative filters;
- any physical reading of the auxiliary sphere as spacetime.

13 Standing results, weak points, and deferred work

What remains standing:

- phase/time separation;
- absolute/relative entropy bookkeeping;

- monotone floor with relative reset;
- apparent-Hubble capacity as a mainline diagnostic;
- failure-aware classification;
- perturbative memory/regeneration language.

What remains weak or unresolved:

- N_{acc} is not microscopically counted;
- Γ_S is not derived;
- q_{crit} is phenomenological;
- k_c, p, and P_ξ are not derived;
- no action, field equation, bounce, or cosmological solution is provided;
- no observational inference is made.

Deferred work would require a real physical route: a sharper causal-patch construction, an entropy-production model, a restart threshold mechanism, perturbation matching from dynamics rather than ansatz, and eventually comparison with data. The single most important route is the derivation of $T_R(k)$ from a concrete toy transition or bounce-like dynamics. Without such an input, the perturbative sector remains a classifier, not a prediction. None of those are claimed here.

14 Provisional conclusion

The v1.3 update reduces SCT rather than expanding it. The resulting framework is a controlled toy grammar:

$$\begin{aligned} \phi \in S^1, \qquad & \tau \in \mathbb{R}, \\ S_{\mathrm{abs}} = S_{\mathrm{min}}(n) + \Delta S_n, \qquad & S_{\mathrm{min}}(n+1) \geq S_{\mathrm{min}}(n), \\ \widehat{S_{\mathrm{cap}}} = E^{-2}, \qquad & q_{\mathrm{eff}} = \frac{\Delta S}{S_{\mathrm{cap}}}, \\ \delta_{n+1}(k) = T_R(k)\delta_n(k) + \xi_n(k). & \end{aligned}$$

Its current value is conceptual and methodological: it clarifies how a relative restart can be represented without absolute entropy erasure, and it records where the idea fails. It remains a toy/informational framework, not a physical cosmology.

A Legacy v1.2.0 dynamic scan context

The v1.2.0 dynamic phase map was already nontrivial: it contained ok, weak, critical, and failed cases. The v1.3 update does not erase that fact.

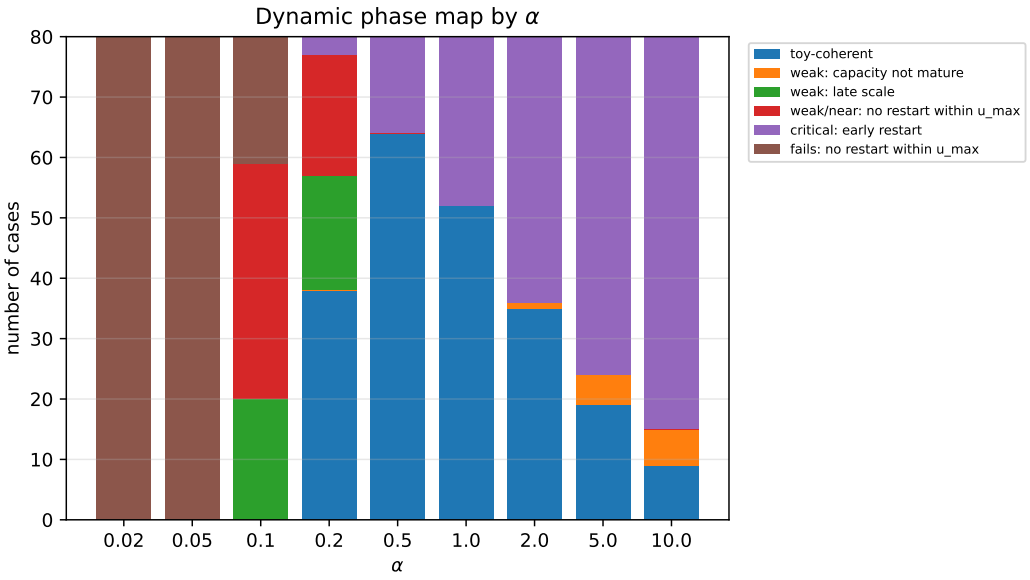


Figure 4: Baseline dependence on α from v1.2.0. Small rates tend to fail by no restart; large rates tend toward premature restart; the middle region is the interesting toy regime.

B Patch-choice sensitivity

The event-horizon diagnostic is retained because it exposes sensitivity. It is not selected as mainline because choosing it solely for better counts would violate the methodological rule.

C Source-template failures

The matter-window, radiation-window, matter–Λ transition, and capacity-growth templates are retained as negative controls. Their failure is part of the audit result.

D Perturbative failure modes

Perturbative matching failure modes include memory erased at all scales, no large-scale transmission, suspicious memory dominance, and small-scale memory dominance. A clean relative restart cannot simply preserve everything; nor can it erase all memory if the goal is partial continuity.

E Selected references

1. R. C. Tolman, *Relativity, Thermodynamics and Cosmology*, Oxford University Press, 1934.

2. P. J. Steinhardt and N. Turok, *A cyclic model of the universe*, Science 296, 1436–1439, 2002.

3. P. J. Steinhardt and N. Turok, *Endless Universe: Beyond the Big Bang*, Doubleday, 2007.

4. A. Ijjas and P. J. Steinhardt, *A new kind of cyclic universe*, Physics Letters B 795, 666–672, 2019.

5. R. Penrose, *Cycles of Time: An Extraordinary New View of the Universe*, Bodley Head, 2010.

6. R. Bousso, *A covariant entropy conjecture*, Journal of High Energy Physics 1999(07), 004, 1999.

7. R. Bousso, *The holographic principle*, Reviews of Modern Physics 74, 825–874, 2002.

8. W. Fischler and L. Susskind, *Holography and cosmology*, arXiv:hep-th/9806039, 1998.

9. R.-G. Cai and S. P. Kim, *First law of thermodynamics and Friedmann equations of Friedmann-Robertson-Walker universe*, Journal of High Energy Physics 2005(02), 050, 2005.

10. G. W. Gibbons and S. W. Hawking, *Cosmological event horizons, thermodynamics, and particle creation*, Physical Review D 15, 2738–2751, 1977.

11. J. D. Bekenstein, *Black holes and entropy*, Physical Review D 7, 2333–2346, 1973.

12. J. M. Bardeen, B. Carter, and S. W. Hawking, *The four laws of black hole mechanics*, Communications in Mathematical Physics 31, 161–170, 1973.

13. J.-L. Lehners, *Ekpyrotic and cyclic cosmology*, Physics Reports 465, 223–263, 2008.